

### 3.3 Database

In this part you will design a database that can save the data required by the project.

The design will begin with a UML conceptual model of the database including all classes, associations, and cardinalities.

The resulting database will later be used as the backend to the recycling center website.

As many recycling centers will be distributed throughout the country, every center has its own code. Address and opening times must be saved in the database.

Only registered clients can bring their waste for recycling. It will be enough to save the username, name, email, password and unique random alphanumeric code(string) which is generated when the user registers for the first time. Every user has their own unique username which is to be used as primary key. Here is an example of a user entry:

| Username | Name           | Email                                            | Type (opt.) | RandomCode | Password   | Fav. Center |
|----------|----------------|--------------------------------------------------|-------------|------------|------------|-------------|
| jschmit  | Jos Schmit     | <a href="mailto:jschmit@pt.lu">jschmit@pt.lu</a> | user        | 345fgt678  | xxxxxxxxxx | RCLUX       |
| dmueller | Daniel Mueller | dmueller@admin.net                               | admin       | fgt66781   | xxxxxxxxxx |             |

Every Recycling center must be recorded in the database with its full address and opening times. Every center has a unique identifier. Here an example of a recycling center entry:

| Center code | Address          | Postal code | Town       | Open | Close |
|-------------|------------------|-------------|------------|------|-------|
| RCLUX       | 1, rue de Beggen | 112345      | Luxembourg | 8:30 | 18:00 |

It must be possible that the user's favorite recycling center is recorded in the database.

Every measurement station where a user weighs its waste must also be recorded in the database. A station has a unique number and a description. A measurement station belongs to a specific recycling center. This must be traceable.e...

As the price the user needs to pay, depends on the weight and the type of waste, all types must be recorded in the database. Every waste type has a unique code by which it is identified. Here an example of a waste entry:

| WasteCode | Description       | Price |
|-----------|-------------------|-------|
| PLA       | Plastic           | 0.2€  |
| BIO       | Kitchen Bio Waste | 0.1€  |

When a user weighs its waste, it must be possible to save not only the weighted value, but also the type of waste, the measurement station, and the recycling center in which the action took place. Here is an example of such a measurement:

|                                               |             |            |                   |            |
|-----------------------------------------------|-------------|------------|-------------------|------------|
| Recycling Centre: Luxembourg-City             | Code: RCLUX |            |                   |            |
| User: Jos Schmit                              |             |            |                   |            |
| Measurement Station No 145                    | Measure No  | Weight(kg) | Type              | Date       |
| Description: Hard plastic and glass materials | 451         | 25         | Plastic           | 05/05/2023 |
|                                               | 452         | 30         | Flat Glass        | 05/05/2023 |
| Measurement Station No 146                    | 4625        | 15         | Kitchen bio waste | 05/05/2023 |
| Description: Biological                       | 4626        | 30         | Grass             | 05/05/2023 |

The user's measurements of the day will then result in an invoice where the prices for the individual measurements as well as the total price will be listed. Below an example of such an invoice:

|                                   |            |                  |                   |       |
|-----------------------------------|------------|------------------|-------------------|-------|
| Invoice No: 2657                  |            | Date: 06/05/2023 |                   |       |
| User: Jos Schmit                  |            |                  |                   |       |
| Recycling Centre: Luxembourg-City |            |                  |                   |       |
| Summary                           | Measure No | Weight(kg)       | Type              | Price |
|                                   | 451        | 25               | Plastic           | 5.0€  |
|                                   | 452        | 30               | Flat Glass        | 7.0€  |
|                                   | 4625       | 15               | Kitchen bio waste | 1.5€  |
|                                   | 4626       | 30               | Grass             | 4.0€  |
| TOTAL:                            |            | 17.5€            |                   |       |

The operators of the recycling center would also like to know if an invoice was paid or not.